

YEAR 9 SCIENCE (MODIFIED)
CHEMICAL SCIENCES
TEST 1 - ATOMIC STRUCTURE, RADIATION

NAME: SOLUTIONS

CLASS: _____

Mark: 15

All matter is made of atoms that are composed of protons, neutrons and electrons; natural radioactivity arises from the decay of nuclei in atoms.

Multiple Choice Write the answer to each question in the appropriate box at right.

1. γ , β , α , B

IN ORDER, the above symbols mean:

- A Gamma, Beta, Alpha, Boron.
 B Gamma, Boron, Alpha, Beta.
 C Alpha, Beta, Gamma, Beryllium.
 D Alpha, Beryllium, Gamma, Beta.

2. The nucleus of which atom is identical to an **alpha particle**?

- A H - Hydrogen
 B He - Helium
 C Li - Lithium
 D Ti - Titanium

3. Which of the following statements about atoms is **true**?

- A An atom is mostly empty space.
 B Neutrons and protons are found in the nucleus with the electrons.
 C Electrons circle the nucleus all the time.
 D The number of neutrons found in the nucleus of an atom is always the same.

4. The **mass number** of an element is the total number of:

- A electrons in an atom.
 B protons in an atom.
 C protons and electrons in an atom.
 D protons and neutrons in an atom.

5. The **atomic number** of an element is the total number of:

- A electrons in an atom.
 B protons in an atom.
 C protons and electrons in an atom.
 D protons and neutrons in an atom.

QUESTION	ANSWER
1	A
2	B
3	A
4	D
5	B
6	B
7	A
8	A
9	D
10	B

6. A charged atom is called:
- A an isotope.
 - B an ion.
 - C a proton.
 - D radioactive.
7. Beta particles are:
- A fast-moving electrons.
 - B positively charged.
 - C heavier than an alpha particle.
 - D unable to penetrate human skin.
8. One of the particles inside the atom is very much lighter than the others. It is the:
- A electron.
 - B neutron.
 - C nucleus.
 - D proton.
9. The nucleus of the atom contains:
- A only neutrons.
 - B only protons.
 - C electrons and neutrons.
 - D neutrons and protons.
10. Different isotopes of the same element contain different numbers of:
- A electrons.
 - B neutrons.
 - C protons.
 - D nuclei.

Short Answer Write the answer to the questions in the spaces provided.

1. Complete the following table.

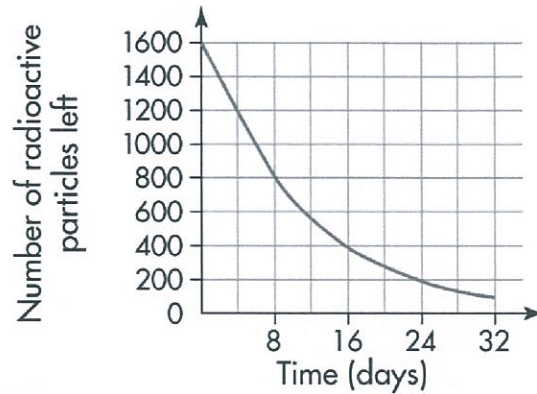
ELEMENT	PROTONS	NEUTRONS	ELECTRONS
${}^{19}_{9}\text{F}$	9	10	9
${}^{39}_{19}\text{K}$	19	20	19

(1)

(1)

(2)

2. The graph below shows the radioactive decay of iodine-131.



What is the half-life of iodine-131?

8 days (1)

(1)

3. List **ONE** major long-term effects on humans caused by cell damage resulting from exposure to large amounts of nuclear radiation.

- Cancers - various types.
- Damage to DNA.
- Permanent cell damage - burns, etc.

[Any 1 reasonable effect]

(1)

4. Describe **ONE** uses of nuclear radiation in medicine and/or industry.

- Nuclear power.
- Medicine - tracers, radiation treatment.
- Industry - testing welds, tracing fluid movement.
- Smoke detectors.

(1)

[Any 1 reasonable effect.]

